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Epigenetics

Epigenetics:

- How Env influences on development event can affect the phenotype.
- Alteration of gene without changes in DNA seq itself, which may lead to functional changes.
- Chemical modifications in the DNA or the histones instead

↳ **Mitotic Stability:**

↳ Epigenetics stable transmitted to next gen.

↳ **Meiotic Stability:**

↳ next gen

↳ reprograming in gametes!

→ DNA seq is the same!

→ eg: tissue specific

Packing of DNA

Nucleosome: 146 bp DNA wrapped around a core histone complex!

↳ stable 1st structure of chromatin

↳ then higher order folding

GCB

Chromatin

Chromatin in the nucleus is organized into areas with different condensation states.

→ Euchromatin : less densely packed or less electron dense (EM) domains in the interphase nucleus.
Histones are marked by euchromatic modifications

→ Heterochromatin : More densely packed, electron dense and intensively stained by Feulgen in the interphase nucleus.
Histones contain specific heterochromatic modifications.

Interphase vs Prometaphase

→ Interphase nucleus : chromosomes are difficult to distinguish from each other. They do occupy a discrete space inside a nucleus

→ Prometaphase nucleus : chromosome territories transform into highly condensed chromosomes, so much that can be easily distinguished.

Nucleosome : basic unit of chromatin

→ Nucleosome consists of a histone octamer and DNA wrapped around the histone octamer.

→ DNA 160 base pairs

→ Histone tails can be chemically modified and alter the histones' chemical properties

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Histone Modification

-> Mods are established/removed when histones are in nucleosomes.
   ↳ by directly or exchanging histones.
-> HM are associated with a chromatin structure
-> Controlled by specific enzymes in forward and reverse reaction
   eg: Methylation, Acetylation, Phosphorylation
-> Effect of altered acetylation:
   opening/closing of chromatin for transcription

Readers of histone modifications

-> HMs recruit proteins to chromatin
-> Demethylases often read other HMs to be targeted
-> Repressor & Activator recognize HMs

Nucleosome exchange in chromatin

Regulation of histones density in chromatin:

I) Histones are assembled by general histone chaperones into newly formed nucleosomes
II) Single ~~modification~~ modified histone of dimers/tetramers can be exchanged by specific chaperones against unmodified histones or histone variants
III) Nucleosomes can be moved along the DNA to release or hide certain DNA seq's for recognition = Nucleosome Remodeling

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Nucleosome Assembly

→ deposition of histones in chromatin is catalyzed by specific enzymes (histone chaperones)
↳ guide Histones complexe to part of the DNA
↳ important for DNA-replication / DNA-repair

→ How are HMs inherited when new nucleosome is formed?
↳ new nucleosome is unmodified!

→ Nucleosome dynamics & epigenetic calvd:
→ opened → active ; → closed → deactive

Histone Variants

→ Transcription linked to changes in the nucleosome/histone composition

Methods for Analysis of HM

→ Analysis of mod histones in cells: → Western-Blot
→ Indirect Immunofluorescence

→ Qualitative & quantitative analysis of HM in cells:
→ ChIP-seq
+
↳ Chromatin Immuno Precipitation: analysis at specific regions
↳ Technik um Bindungsstellen von TFs auf der DNA zu detektieren.

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Peak Calling

↳ identify enriched (signal) regions in ChIP-seq.

eg: MACS2

→ Quality Measures:

↳ Fraction of reads in peaks (FRiP) ⇒ depends on data type.

↳ PCR Bottleneck Coefficient (PBC) is a measure of library complexity

↳ Relative Strand cross-corelation

0: no signal, >1: highly enriched experiments

↳ Normalized Strand cross-corelation NSC

>1.1: more enrichment, 1 = (low) no enrichment

↳ IDR: Irreproducible Discovery Rate:

low significance → low consistency

most significant peaks ⇒ high consistency between replicates.

↳ Measures consistency between replicates

in high-throughput experiments ⇒ Identify outliers

MACS2

→ Most widely used peak caller.

→ Identifies genome wide locations of TF binding, HM or NFRs from ChIP-seq or ATAC-seq data

→ Can be used without control (not with ChipSeq)

→ Controls eliminates bias

→ Can call narrow & broad peaks

QSB

Peak Calling with MACS2

Step I : Estimate fragment length d and adjust read position.

Step II : Identify local noise

↳ Estimate λ_{local} of Poisson Distribution

→ local characteristics of the genome can lead to a bias in the # of reads being mapped.

↳ use a dynamic λ_{local} : $\lambda_{local} = \max(\lambda_{BG}, \lambda_{1k}, \lambda_{5k}, \lambda_{10k})$

↳ over entire genome

↳ reduces the influence of local biases and is robust against occasional low tag counts in small regions

Step III : Identify enriched (peak) region

↳ regions s.t. : p-value < PVALUE

↳ determine summit position within enriched region as max density

Step IV : Estimate FDR

Reads to Peaks

→ Reads from both strands can be extended to the length of d (Fragment length)

→ Reads can be shifted by d/2

⇒ Different Peaks Different Shape

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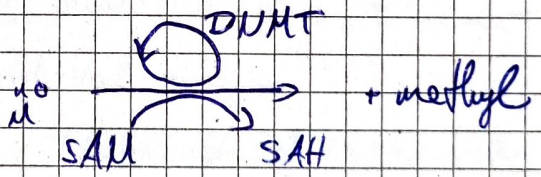
DNA methylation

-> maintenance of genome stability and integrity
-> Control of developmental processes (diff cells)
-> Control of tissue specific expression
-> Imprinting:
  -> Epigenetic marking of a locus on the basis of parental origin
  -> Results in mono-allelic gene expression
-> X chromosome inactivation:
  -> Gene expression on one of the female X-chromosome is inactivated
  -> Calico cats

-> & cancer:
  Hypermethylation (up) & Hypomethylation (down) <=> Cancer
  L> of tumor suppressor gene L> of oncogene promoters

-> Therapy: Epigenetic modifier inhibitors

-> Enzyme Reaction



L> does not affect base pairing

-> Base specific:
  -> occurs in a seq context and usually is symmetrical on both strands
  -> is almost exclusively confined to the CpG seq context

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CpG islands

-> not equally distributed

-> Dense Regions of CG & G => CpG Islands

-> are often associated with gene promoters

-> Gene expression is correlated with high levels of gene body (genic) methylation.

-> CpG-poor region, with the exception of enhancers, are typically methylated in normal cells.

-> Similarly, CpG-poor promoters are silenced by DNA-methylation & exhibit a closed chromatin structure.

Why Study Methylation

-> epigenetic mark with mitotic inheritance

-> For: Genomic imprinting, Inflammation

Workflow: Analyzing BS-data

-> Process: Bisulfite-seq data

unmethylated C -> U
methylated C -> C
PCR: U -> T
Discriminate

-> Data visualization and statistical analysis

-> Downstream analysis: Identification of Differentially Methylated regions (DMRs),

Global

-> Methylation status of all individual C's in genome

-> Find the max DMR with at most k exceptions:

ex ≤ 2, $p \leq 0,05$ good, ~~$p = 0,0$~~, window size = 5

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Downstream analysis & integrative tools

Downstream Functional Analysis:

-> primary analysis for epigenetic assay is done
↳ Out : set of peak locations, CpG islands

↳ functional analysis by comparing :
- Different groups, Different region from same dataset
↳ use public DBs
- eg: IHEC,
provides standardized reference epigenomes for a variety of normal & disease tissues

Epigenome-wide Association Study (EWAS)

An examination of a genome-wide set of quantifiable epigenetic marks, such as DNA methylation, in different individuals to derive associations between epigenetic variations and a particular identifiable phenotype.

GSS-Gr 10

(1)

a) Epigenetics:
Das Fachgebiet der Biologie, die sich mit der Frage befasst, welche Faktoren die Aktivität eines Gens und damit die Entwicklung der Zelle entweder festlegen.
Chemical modifications in DNA or histones.

b) Chromatin
Mixture of DNA and Proteins that form the chromosomes in the cell of higher organisms.
DNA-Strang auf Histones aufgewickelt

c) Histone
Protein that provides structural support for a chromosome.
DNA wraps around complexes of histons → more compact + regulation

d) Histone Modification
Methylation, Acetylation, Phosphorylation
tiny covalent addition of a chemical group

e) CpG islands.
regions in DNA where C followed by G occurs frequently. constantly methylated!

GASS

④ Histone Modification

a)

H3K4me1 = Histone 3 monomethylated at lysine 4
H3K9me3 = Histone 3 trimethylated at lysine 9
H4K20me2 = Histone 4 dimethylated at lysine 20
H3K4Ac = Histone 3 acetylated at lysine 4

b)

Euchromatin is defined as active chromatin with a low level of compaction in the interphase nucleus

Heterochromatin is defined as chromatin which is more closely packed than euchromatin

⑤ Peak Calling

a) MACS2:

→ should be used with an Input seq (and recommended) when calling ChIP-peaks
→ can call peaks on ATAC-seq data
→ can call peaks on TF-ChIP-data
→ can call narrow peaks
→ can call broad peaks

b) MACS infer p-Value

→ computes p-value based on a local Poisson Distribution of the BG

c) IDR

→ IDR measures consistency between replicates in high-throughput experiments

GSB - Ex 10

(4) DNA methylation

a) CpG islands

-> are stretches of CpGs that are constantly methylated

b) Effect of DNA Methylation

↳ promoter region => decrease gene expression
↳ enhancer region => decrease gene expression
↳ gene body => increase gene expression

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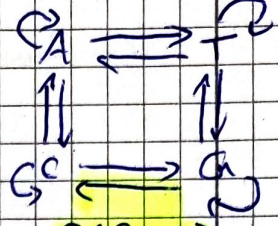
Markov Chain

-> used by Hidden Markov Models

-> Idea: Stochastic process for modeling state changes by knowing a limited history



-> Properties:
- Memoryless
- Homogeneous
- Time-discrete / stationary process
- Transition Matrix

-> Random Variables: $Y = (X_t)$, $X_t \in Q = \{q_{f1}, \ldots, q_n\}$ (f = state)

-> 1st order Markov Chain: $P(X_{t+1} = q_{j(t+1)} \mid X_t = q_{jt})$

-> Transition probabilities: $p_{ij}(t) = T(X_{t+1} = q_j \mid X_t = q_i)$

-> Transitionmatrix TM ↙

-> n^{th} order: $P(X_{t+1} = q_{j(t+1)} \mid X_t = q_{jt}, \ldots, X_{t-n+1} = q_{j(t-n+1)})$

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Hidden Markov Chain

→ Generalization of MC

→ 2 types of probabilities:

→ States of the model (hidden state), P(transition)

→ Observable events (observable state), P(emission)

→ P depends only on previous state

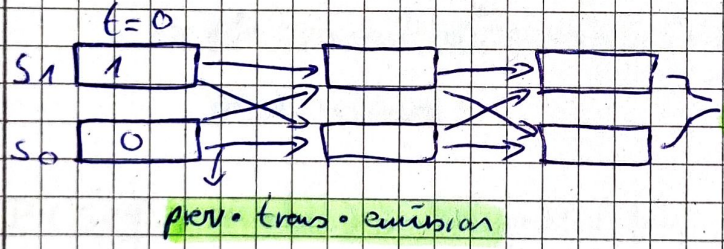
→ eg: Coding / non-coding regions in DNA determination, Gene prediction

Problems

I) Evaluation: Given a model and an output sequence, what is the P that the model generated that output?

→ all possible paths

→ Forward Algorithm: Given Model. $S_0 \rightleftarrows S_1$ (0,3 / 0,7 / 0,8 / 0,2)



$$P(\underbrace{X_1 \dots X_L}_{\text{seq}}, \underbrace{q_0 \dots q_{L+1}}_{\text{path}} \mid \text{model}) = T(q_{L+1} \mid q_L) \leftrightarrow \sum_{i=1}^{N} T(q_i \mid q_{i-1})\, P(X_{L(i)} \mid q_i)$$

$$= \sum_{\text{paths}} P(X_1 \dots X_L, q_0 \dots q_{L+1} \mid \text{model})$$

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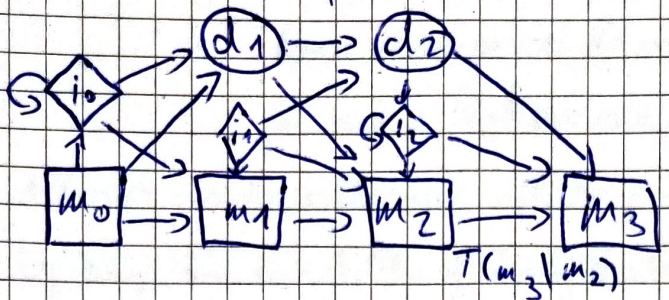
HMMs

→ for protein seq generation



M = 2

m = match state (position in a protein) (MSA)

i = insert state

d = delete state

P(x|q) : that letter x in state q

T(r|q) : transition to state r from q

→ Markov assumption:

→ the next state depends only on the current state

→ Stationary assumption:

→ state transition probabilities are independent on time

→ Output independence assumption:

→ the current observation is independent of previous observations

P(seq's| model): The probability of a set of training seq's s(1), ..., s(n)

Ch 5 B

Problems

III) Learning: Given a model and a set of observed seq's, how do we set the models parameters so that it has a high probability of generating those seq's?

→ Allows us to determine all the probabilities in an HMM by using an ensemble of train data.

EM - Algorithm: Forward-Backward-algorithm

→ In: Topology of HMM & Data

→ Maximum Likelihood (ML) of the model:

In: $s(1), .., s(n)$ and find

$$P(\text{seq's} \mid \text{model}) = \prod_{j=1}^{n} P(s(j) \mid \text{model}) \rightarrow \max$$

↳ Negative log Likelihood: NLL-score $= -\log P(\text{seq's} \mid \text{model})$

→ Maximum a posteriori (MAP) of the model:

$$P(\text{model} \mid \text{seq's}) = \frac{P(\text{seq's} \mid \text{model})\, P(\text{model})}{P(\text{seq's})}$$

Bestone:

$$P(\text{seq's} \mid \text{model})\, P(\text{model}) \rightarrow \max$$

→ Idea: Iterative adaption of the model to fit the data

II) Initialize: $T(r|q)$ & $P(x|q)$ $\forall x, q, r$: prior Knowledge

III) Get new estimate of: $T(r|q)$: count # of times a transition in model from state q to r

$P(x|q)$: count # of times the AA x is aligned to the state q

IV) Replace estimations

V) Repeate II & III until convergence

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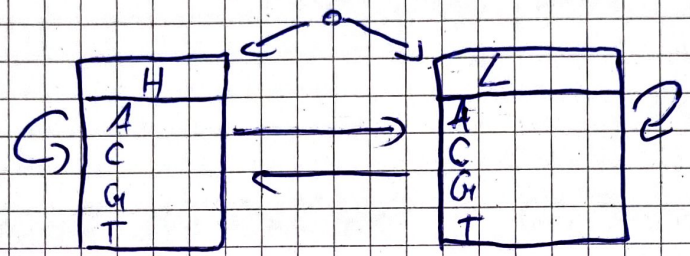
Problems

III) Decoding : Given a model and an output seq, what is the most likely state seq through the model that generated the output ?

→ Gives us a way to match up an observed seq with the states in the model ⇒ Viterbi Alg

Intergenic, start/stop, coding state

eg :



eg : $P_H(G,1) = -1 - 1,7$

$P_H(G,2) = -1,7 + \max($
$P_H(G,1) + P_{HH},$
$P_L(G,1) + P_{LH})$

↳ Model can be used to predict the region of coding DNA from a given seq.

→ Dynamical programing algorithm that allows us to compute the most probable path.

P : → Most probable path ending in state k with observation i :

$$P_l(i,x) = e_l(i) \max_k (P_k(j,x-1) \cdot P_{kl})$$

$e_l(i)$: P of observe i in state l ← $P_k(j,x-1)$: Recursion ← P_{kl} : P of transition from k to l

→ use log to use sums not products

→ backtrack path from last max only to start !

TOOL = HMMER

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Summary

I) Forward / Backward alg: Get the probability of each state at each position.
Sum over all joint probabilities.

II) Expectation / Maximization: Refine the parameters of the model using Data

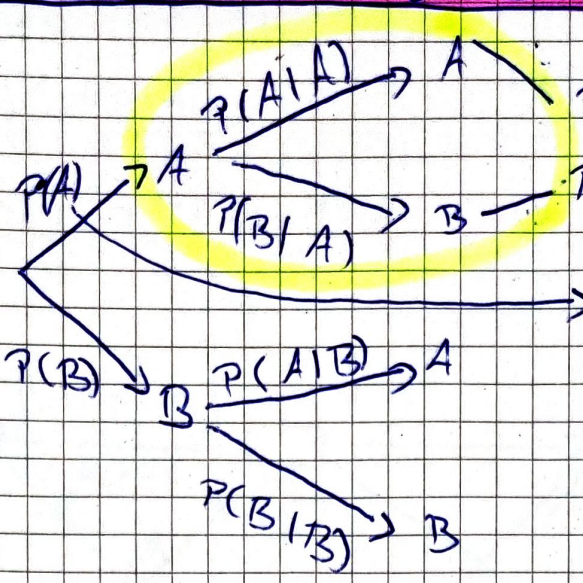
III) Viterbi algorithm: Get the optimal state pathway.
Max joint probability

Advantages: (+) Statistics (+) Transparency
(+) Modularity (+) Prior Knowledge

Disadvantages: (-) State independence (-) Local Max's
(-) Over-fitting (-) Speed

Good Solution:
MSA Protein classification
Seq clustering Protein Domain identification

GMSB - Ex 11



$P(AA) = P(A)P(A|A)$

$P(AB) = P(A)P(B|A)$

$P(A) = \sum P(AX)$

(1) Markov-Chain

$P(XX)$ are given

a) 1st Order: We need $P(Y|X) = \frac{P(XY)}{P(X)}$

↳ $P(X) = \sum P(XY)$

$P(A) = P(AA) + P(AT) + P(AC) + P(AG) = 0{,}33$

$P(T) = 0{,}29$, $P(C) = 0{,}2\,?$, $P(G) = 0{,}18$

	A	T	C	G
A	0,27			
T	0,31			
C				
G				

$P(A|A) = \frac{P(AA)}{P(A)} = 0{,}27$

$P(A|T) = \frac{P(TA)}{P(T)} = 0{,}31$

b) $P(Z) = P(Z_n|Z_{n-1}) \cdot \ldots \cdot P(X_2|X_1)P(X_1)$

(2) Viterbi Alg

$P(S_1) = 0, P(S_2) = 1$